

A mod-four obstruction to extending the four six-face seeds

Research note: exact certificate and independent enumeration audit

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For a word prefix let c_a count occurrences of letter a , and let p_{ab} count scattered ordered pairs ab . The eight canonical three-wise fair words on four letters with six occurrences of each letter, found by exhaustive enumeration, are

w	t
012332102103301231022013	2
012332102103301231200213	2
012332102130031231022013	2
012332102130031231200213	2
012332103102201321033012	3
012332103102201321300312	3
012332103120021321033012	3
012332103120021321300312	3

Here canonical means that labels are assigned in order of first occurrence.

Proposition 1. *Inserting an additional M -face die into any of these eight words can preserve three-wise fairness only if $4 \mid M$.*

Proof. For the parameter t in the table define the prefix statistic

$$L = c_1 - p_{01} - p_{21} - p_{31} - 2p_{t1}.$$

For every prefix of each corresponding word, L is divisible by four. Indeed, it changes only when a letter 1 is appended. Its values just after the six occurrences of that letter are, in all eight cases,

$$0, -8, -20, -36, -56, -84.$$

Let z_g be the number of new letters inserted in gap g , including the two exterior gaps. Three-wise fairness implies pairwise fairness, so the pair consisting of old letter 1 and the new letter gives

$$\sum_g z_g c_1(g) = 3M.$$

For each $a \neq 1$, the order old a , old 1, new letter must have probability $1/6$. Since the old dice each have six faces, this says

$$\sum_g z_g p_{a1}(g) = 6M.$$

Consequently

$$0 \equiv \sum_g z_g L(g) = 3M - 6M - 6M - 6M - 12M = -27M \pmod{4},$$

which forces $4 \mid M$. The argument even allows signed integer gap multiplicities; no nonnegativity assumption is needed for this obstruction. $\square$

In particular no three-wise fair system of five six-face dice exists, provided the stated enumeration of the four-die seeds is complete. The independent C++ audit `review_small_enumeration.cpp` verifies this completeness by a different implementation:

- It visits all $18!/(6!)^3 = 17\,153\,136$ balanced ternary words, keeps the 2858856 canonical ones, and counts ordered triples by literal enumeration of index triples. Exactly eleven words survive.
- For each of those eleven words it enumerates every nondecreasing six-tuple of insertion gaps directly, without meet-in-the-middle. Among $11\binom{24}{6} = 1\,480\,556$ assignments it finds 32 successful insertions and exactly the eight canonical words above.
- As a redundant check it tests all $8\binom{30}{6} = 4\,750\,200$ six-gap insertions into those eight words and finds none.

The insertion tests impose the pair counts and both orders of old pairs before the new letter. These constraints imply all remaining orders: for distinct old letters a, b and the new letter Z ,

$$\#(aZb) = 6\#(aZ) - \#(abZ) - \#(baZ), \quad \#(Zab) = 6\#(ab) - \#(abZ) - \#(aZb).$$

The analogous two identities with a, b exchanged give all six triples. Every canonical word on one more letter is covered by deleting its last first-occurrence label; the remaining word is itself canonical. Thus neither the pruning in the original enumeration nor a solver's infeasibility report is needed for the independent audit.

Together with the existing explicit twelve-face construction, and the elementary requirement $6 \mid M$ for three-wise fair equal-sided dice, this proves $M_3(5) = 12$.